

Unlocking Customer Insights: A Statistical Investigation

Statistics Mini Project

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I. Introduction:

This project analyzes customer behavior using statistical techniques including descriptive statistics, visualization, and hypothesis testing.

II. Project Objective:

The objective of this project is to analyze customer data to identify patterns in spending, engagement, and demographics, and to support data-driven business decisions.

```
In [1]: import os  
os.getcwd()
```

```
Out[1]: 'C:\\Users\\VORTEX\\OneDrive\\Desktop\\Data Analysis\\Statistics\\Mini Project'
```

III. Data Understanding:

```
In [2]: # Importing Important Libraries  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns  
from scipy import stats
```

```
In [3]: # Loading dataset  
df = pd.read_csv("US_Customer_Insights_Dataset.csv")  
  
print("Loaded Successfully")
```

Loaded Successfully

i) Dataset Overview-

```
In [4]: df.head()
```

Out[4]:

	CustomerID	Name	State	Education	Gender	Age	Married	NumPets	Jo
0	CUST10319	Scott Perez	Florida	High School	Non-Binary	47	Yes	1	9
1	CUST10695	Jennifer Burton	Washington	Master	Male	72	Yes	0	
2	CUST10297	Michelle Rogers	Arizona	Master	Female	40	Yes	2	7
3	CUST10103	Brooke Hendricks	Texas	Master	Male	27	Yes	0	8
4	CUST10219	Karen Johns	Texas	High School	Female	28	Yes	1	1

ii) Data Types and Structure-

In [5]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10675 entries, 0 to 10674
Data columns (total 12 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CustomerID                           10675 non-null  object
1   Name                                 10675 non-null  object
2   State                                10675 non-null  object
3   Education                            10675 non-null  object
4   Gender                               10675 non-null  object
5   Age                                  10675 non-null  int64
6   Married                              10675 non-null  object
7   NumPets                              10675 non-null  int64
8   JoinDate                             10675 non-null  object
9   TransactionDate                       10675 non-null  object
10  MonthlySpend                          10675 non-null  float64
11  DaysSinceLastInteraction               10675 non-null  int64
dtypes: float64(1), int64(3), object(8)
memory usage: 1000.9+ KB
```

In [6]: `df.nunique()`

```
Out[6]: CustomerID      1000
        Name           990
        State           10
        Education         5
        Gender            3
        Age              63
        Married           2
        NumPets           5
        JoinDate         731
        TransactionDate  1605
        MonthlySpend     9843
        DaysSinceLastInteraction 1605
        dtype: int64
```

iii) Checking Missing Values -

```
In [7]: df.isnull().sum()
```

```
Out[7]: CustomerID      0
        Name           0
        State          0
        Education      0
        Gender         0
        Age            0
        Married        0
        NumPets        0
        JoinDate       0
        TransactionDate 0
        MonthlySpend   0
        DaysSinceLastInteraction 0
        dtype: int64
```

iv) Date Engineering-

```
In [8]: df['JoinDate'] = pd.to_datetime(df['JoinDate'], format='%m/%d/%y')
        df['TransactionDate'] = pd.to_datetime(df['TransactionDate'], format='%m/%d/%y')

        print("\nDATA TYPES:\n", df.dtypes)
```

```
DATA TYPES:
CustomerID      object
Name            object
State           object
Education       object
Gender          object
Age            int64
Married         object
NumPets        int64
JoinDate        datetime64[ns]
TransactionDate datetime64[ns]
MonthlySpend    float64
DaysSinceLastInteraction int64
dtype: object
```

IV. Descriptive Statistics:

Descriptive statistics were used to summarize the dataset.

- The mean represents the average value of numerical variables such as age, monthly spending, and customer activity.
- The median indicates the middle value and helps reduce the impact of extreme values.
- The standard deviation measures the variability or spread of the data.
- The mode represents the most frequently occurring value in categorical variables such as gender, education, marital status, and state.

```
In [9]: print("==== DATA SUMMARY ====")

print("\n 📊 Numerical Variables:\n")
print(df.select_dtypes(include=['number']).describe())

print("\n 📄 Categorical Variables:\n")
print(df.select_dtypes(include=['object']).describe())

print("\n ⭐ Modes:\n")
for col in df.select_dtypes(include=['object']).columns:
    print(f"{col}: {df[col].mode()[0]}")
```

==== DATA SUMMARY =====

📊 Numerical Variables:

	Age	NumPets	MonthlySpend	DaysSinceLastInteraction
count	10675.000000	10675.000000	10675.000000	10675.000000
mean	49.474567	1.340515	331.610315	538.469883
std	18.221365	1.150849	225.799253	398.766747
min	18.000000	0.000000	3.890000	1.000000
25%	35.000000	0.000000	165.495000	218.000000
50%	49.000000	1.000000	282.110000	445.000000
75%	66.000000	2.000000	443.255000	788.500000
max	80.000000	4.000000	1740.420000	1791.000000

📄 Categorical Variables:

	CustomerID	Name	State	Education	Gender	Married
count	10675	10675	10675	10675	10675	10675
unique	1000	990	10	5	3	2
top	CUST10798	Joseph McBride	California	Master	Male	No
freq	24	29	1180	2269	3791	5583

⭐ Modes:

CustomerID: CUST10798
 Name: Joseph McBride
 State: California
 Education: Master
 Gender: Male
 Married: No

V. Data Visualization:

a) Histogram (Age, Monthly Spend)

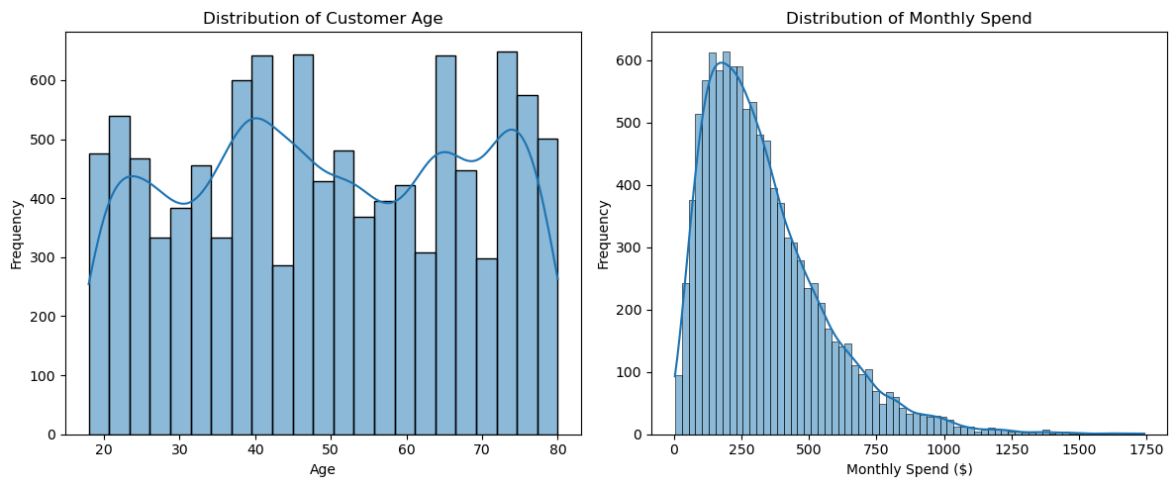
```
In [10]: plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
sns.histplot(df['Age'], kde=True)
plt.title("Distribution of Customer Age")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.subplot(1,2,2)
sns.histplot(df['MonthlySpend'], kde=True)
```

```
plt.title("Distribution of Monthly Spend")
plt.xlabel("Monthly Spend ($)")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```



Insights-

- Age is fairly normally distributed.
- MonthlySpend is right-skewed, indicating presence of high-value customers.

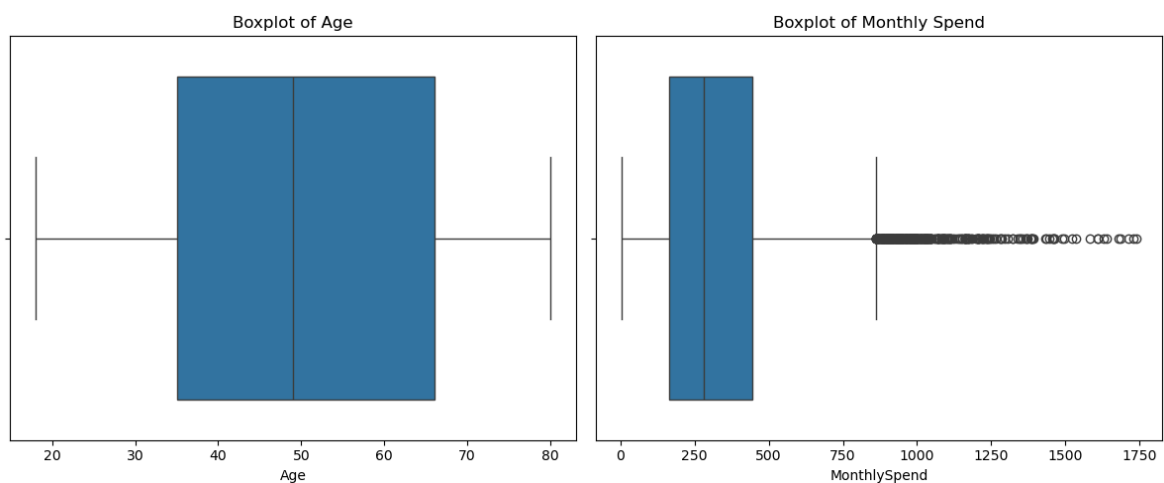
b) Boxplots (Age, Monthly Spend)

```
In [11]: plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
sns.boxplot(x=df['Age'])
plt.title("Boxplot of Age")

plt.subplot(1,2,2)
sns.boxplot(x=df['MonthlySpend'])
plt.title("Boxplot of Monthly Spend")

plt.tight_layout()
plt.show()
```



Insights-

- MonthlySpend shows outliers, confirming high-spending customers.

c) Bar Charts (Gender, Education, State)

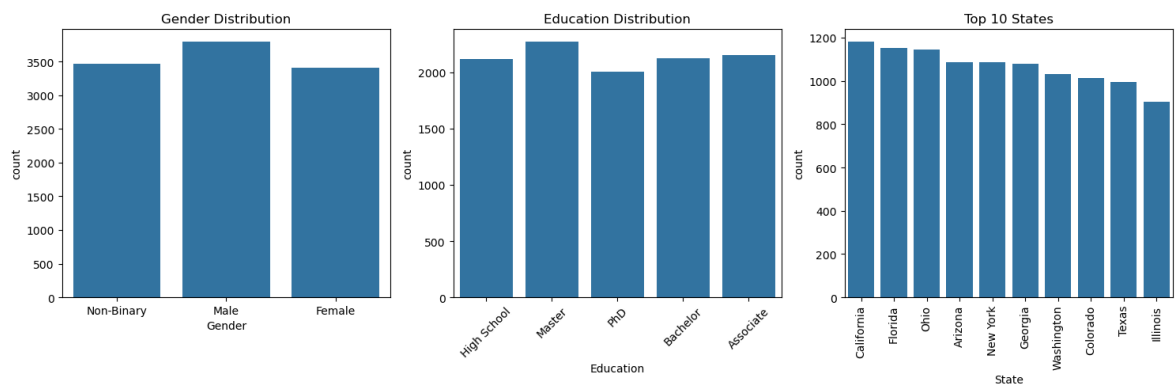
```
In [12]: plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
sns.countplot(x='Gender', data=df)
plt.title("Gender Distribution")

plt.subplot(1,3,2)
sns.countplot(x='Education', data=df)
plt.xticks(rotation=45)
plt.title("Education Distribution")

plt.subplot(1,3,3)
sns.countplot(x='State', data=df, order=df['State'].value_counts().index[:10])
plt.xticks(rotation=90)
plt.title("Top 10 States")

plt.tight_layout()
plt.show()
```

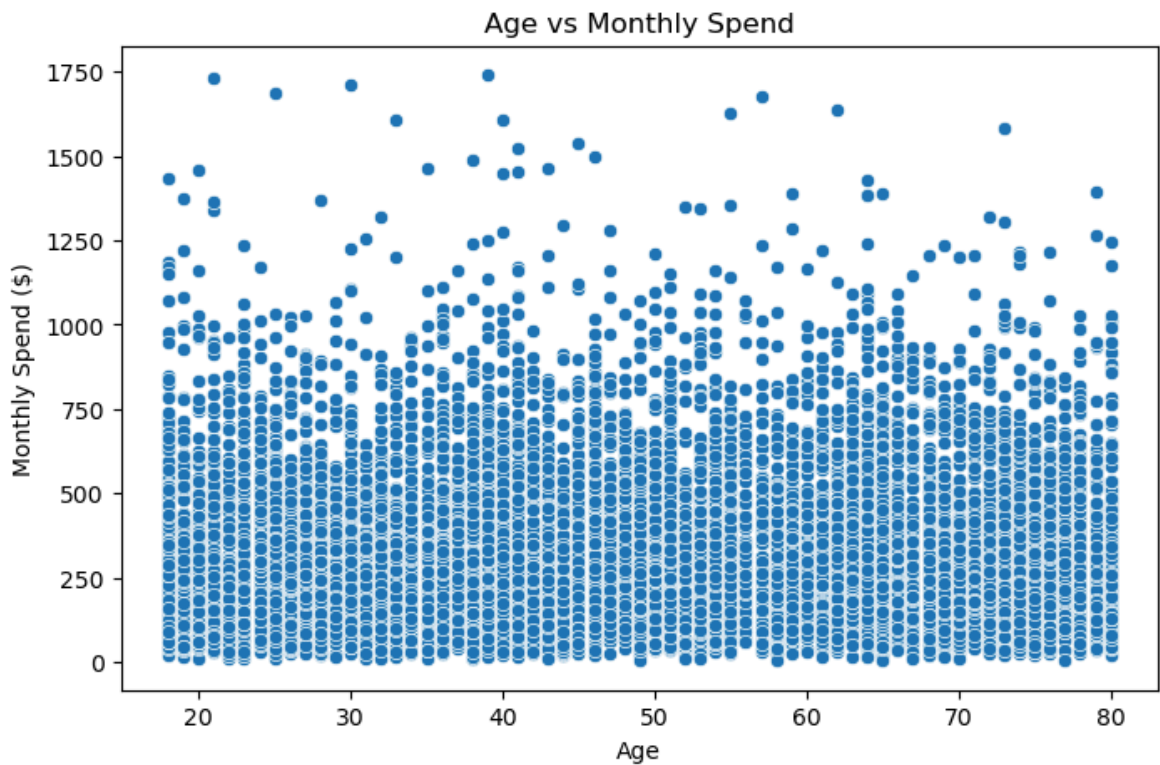


Insights-

- Gender and education are fairly evenly distributed.
- Some states have slightly higher customer concentration.

d) Scatter Plot (Age vs Monthly Spend)

```
In [13]: plt.figure(figsize=(8,5))
sns.scatterplot(x='Age', y='MonthlySpend', data=df)
plt.title("Age vs Monthly Spend")
plt.xlabel("Age")
plt.ylabel("Monthly Spend ($)")
plt.show()
```



Insights-

- No clear trend → Age does not strongly influence spending

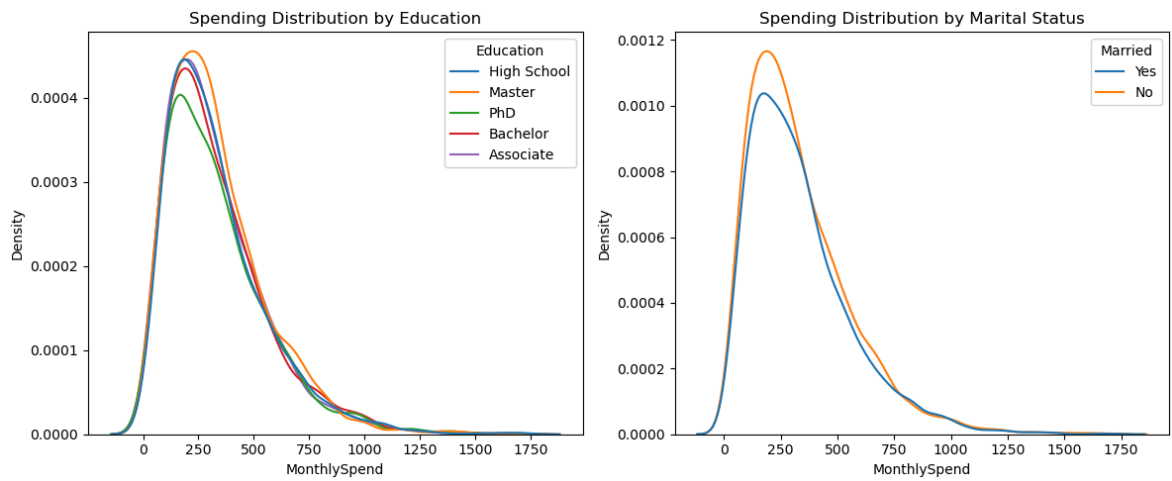
e) KDE Plot (Spending Distribution)

```
In [14]: plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
sns.kdeplot(data=df, x='MonthlySpend', hue='Education')
plt.title("Spending Distribution by Education")

plt.subplot(1,2,2)
sns.kdeplot(data=df, x='MonthlySpend', hue='Married')
plt.title("Spending Distribution by Marital Status")

plt.tight_layout()
plt.show()
```



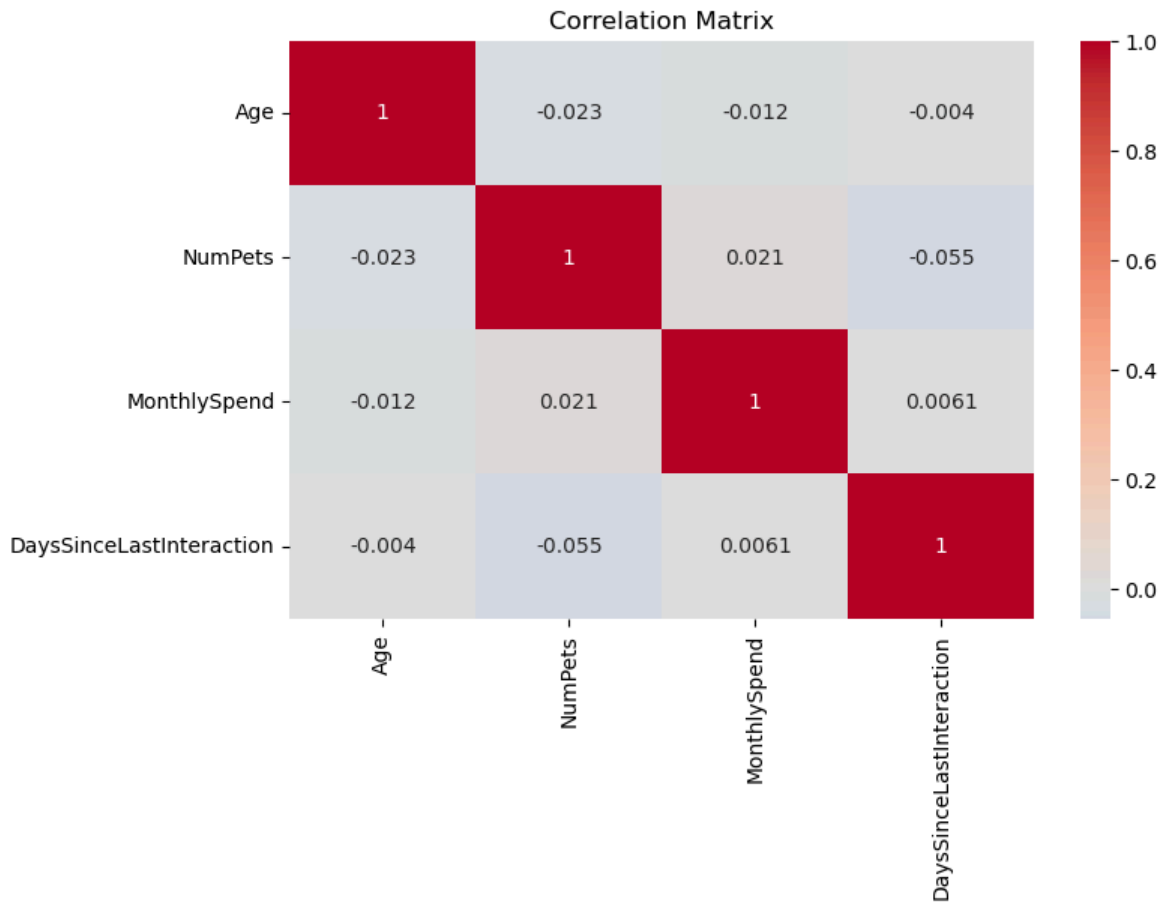
Insights-

- Spending patterns are similar across education & marital groups

VI. Bivariate Analysis:

1. Correlation Matrix

```
In [15]: plt.figure(figsize=(8,5))
corr = df[['Age', 'NumPets', 'MonthlySpend', 'DaysSinceLastInteraction']].corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', center=0)
plt.title("Correlation Matrix")
plt.show()
```

Insights-

- Correlations are weak - no strong linear relationships.

```
In [16]: corr, p = stats.pearsonr(df['Age'], df['DaysSinceLastInteraction'])

print("Correlation:", corr)
print("P-value:", p)
```

Correlation: -0.003970230104955046

P-value: 0.6816905437300954

2. Crosstab (Gender vs Married)

```
In [17]: pd.crosstab(df['Gender'], df['Married'])
```

```
Out[17]:
```

	Married	No	Yes
Gender			
Female	1797	1616	
Male	1892	1899	
Non-Binary	1894	1577	

3. Grouped Statistics (MonthlySpend by State, Education, Gender)

```
In [18]: df.groupby('State')['MonthlySpend'].mean()
```

```
Out[18]: State
Arizona      341.489135
California    339.183492
Colorado      323.083462
Florida       327.696892
Georgia       328.354648
Illinois      332.589591
New York      332.151244
Ohio          340.187860
Texas         319.506770
Washington    329.444078
Name: MonthlySpend, dtype: float64
```

```
In [19]: df.groupby('Education')['MonthlySpend'].mean()
```

```
Out[19]: Education
Associate     327.884408
Bachelor      331.884753
High School   332.215712
Master        334.252305
PhD           331.690090
Name: MonthlySpend, dtype: float64
```

```
In [20]: df.groupby('Gender')['MonthlySpend'].mean()
```

```
Out[20]: Gender
Female        331.361310
Male          333.174068
Non-Binary    330.147240
Name: MonthlySpend, dtype: float64
```

VII. Hypothesis Testing:

- H0: No significant relationships exist between variables
- H1: Significant relationships exist between variables

1. Age vs MonthlySpend (Correlation Test)

- H0: No correlation between Age and MonthlySpend
- H1: Correlation exists

```
In [21]: corr, p = stats.pearsonr(df['Age'], df['MonthlySpend'])

print(f"Correlation: {corr:.4f}")
print(f"P-value: {p:.4f}")
```

Correlation: -0.0123

P-value: 0.2030

* Correlation Test Results:

- correlation: -0.012
- p-value: 0.203
- Decision: p-value > 0.05 → fail to reject H0

- Conclusion: No significant relationship

2. Age vs Customer DaysSinceLastInteraction (Correlation Test)

- H_0 : There is no correlation between Age and DaysSinceLastInteraction
- H_1 : There is a correlation between Age and DaysSinceLastInteraction

```
In [22]: corr, p = stats.pearsonr(df['Age'], df['DaysSinceLastInteraction'])

print(f"Correlation: {corr:.4f}")
print(f"P-value: {p:.4f}")
```

Correlation: -0.0040

P-value: 0.6817

* Correlation Test Result:

- Correlation: -0.004
- p-value: 0.682

Decision: • Since p-value > 0.05, we fail to reject the null hypothesis

Conclusion: • There is no significant relationship between age and customer activity.

Older customers are not necessarily less active.

3. Gender vs Customer Activity (T-test)

- H_0 : There is no difference in MonthlySpend between males and females
- H_1 : There is a difference

```
In [23]: male = df[df['Gender'] == 'Male']['DaysSinceLastInteraction']
female = df[df['Gender'] == 'Female']['DaysSinceLastInteraction']

stats.ttest_ind(male, female)
```

```
Out[23]: TtestResult(statistic=np.float64(1.6704389444143242), pvalue=np.float64(0.094876002312686), df=np.float64(7202.0))
```

* T-test Results:

- t-statistic: 1.67
- p-value: 0.0949
- Decision: p-value > 0.05 → fail to reject H_0
- Conclusion: No significant difference between groups

4. Education vs MonthlySpend (ANOVA)

- H_0 : Mean MonthlySpend is same across all education levels

- H1: At least one group is different

```
In [24]: groups = [group['MonthlySpend'].values for name, group in df.groupby('Education')
stats.f_oneway(*groups)
```

```
Out[24]: F_onewayResult(statistic=np.float64(0.2288066867370917), pvalue=np.float64(0.922359467759936))
```

*** ANOVA Test Result:**

- F-statistic: 0.23
- p-value: 0.9223
- Decision: p-value > 0.05 → fail to reject H0
- Conclusion: Education level does NOT significantly affect MonthlySpend

5. Categorical Relationship b/w Marital status vs Number of pets (Chi-square)

- H0: No relationship between marital status and number of pets
- H1: There is a relationship

```
In [25]: table = pd.crosstab(df['Married'], df['NumPets'])
stats.chi2_contingency(table)
```

```
Out[25]: Chi2ContingencyResult(statistic=np.float64(177.63953668537033), pvalue=np.float64(2.3957232932397494e-37), dof=4, expected_freq=array([[1592.00487119, 1676.20749415, 1436.15156909, 578.95840749, 299.67765808], [1451.99512881, 1528.79250585, 1309.84843091, 528.04159251, 273.32234192]]))
```

*** Chi-Square Test Result:**

- Chi-square statistic: 177.64
- p-value: 2.39e-37
- Degrees of freedom: 4
- Decision: p-value < 0.05 → reject H0
- Conclusion: There is a significant relationship between marital status and number of pets

VIII. Business Insights:

1. Customer spending is highly variable and right-skewed, indicating the presence of high-value customers who contribute disproportionately to revenue.

2. No significant relationship was found between age and spending, and education level does not significantly affect spending, suggesting that demographic-based targeting may not be effective.

3. Customer engagement (DaysSinceLastInteraction) does not significantly differ across genders, indicating uniform engagement behavior.

4. A strong relationship exists between marital status and number of pets, highlighting an opportunity for niche marketing (e.g., pet-related products).

5. Average spending across states appears relatively similar, indicating broadly consistent customer behavior across regions.

Additional Insights -

- Customers with higher monthly spending form a small but important segment contributing significantly to revenue.
- Since most variables show weak correlations, customer behavior is likely influenced by multiple factors rather than a single demographic variable.
- Customers with longer inactivity periods (high DaysSinceLastInteraction) represent potential churn risk and should be targeted with re-engagement campaigns.
- The presence of variability in spending suggests the opportunity to classify customers into low, medium, and high-value segments.

IX. Recommendations:

- Personalize marketing based on spending and engagement behavior
- Target high-spending customers with premium offers
- Re-engage inactive customers through segmentation strategies
- Use pet ownership and behavioural patterns for targeted campaigns
- Monitor customer engagement regularly to reduce churn

Executive summary:

This analysis used descriptive statistics, visualizations, and hypothesis testing to uncover patterns in customer behavior. Key insights include diverse spending behavior, minimal differences in engagement by gender or geography, and opportunities for pet-based and behavior-driven targeting. Recommendations include segmentation-based marketing campaigns to optimize revenue and re-engagement.

X. Conclusion:

Customer behavior shows high variability in spending. However, no strong statistical relationships were found between age, gender, or geography and key metrics.